EASTERN INTERNATIONAL UNIVERSITY
BECAMEX BUSINESS SCHOOL

DATABASE IMPLEMENTATION AND SALES DATA ANALYSIS USING THE CHINOOK DATABASE

Course: MIS 443_Business Data Management

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I. Introduction & Business Problem Statement

1. The Business Problem

Chinook is a global digital media store. Recently, the executive team has noticed fluctuations in yearly revenue and wants to understand the underlying causes. Furthermore, on July 1, 2023, the company rolled out a major User Interface (UI) update. Management needs to know if this update helped or hurt sales, which regional markets are the most profitable, which genres should be prioritized, and how well the sales team is performing.

2. Methodology & Tooling (How we solved it).

To extract actionable business insights from raw data, we utilized a two-step technical approach:

- **PostgreSQL (Data Extraction & Processing):** SQL was strictly used as the backend engine to query the raw database, join multiple relational tables (e.g., matching customers to invoices), handle missing values, and aggregate complex metrics (using CTEs and date intervals for Before & After analysis).
- **Python & Jupyter Notebook (Data Visualization):** Python (using pandas and matplotlib) was used as the front-end analytical tool. We connected Python directly to PostgreSQL to fetch the aggregated SQL results and generate visual charts. This visual approach translates raw database numbers into easy-to-understand trends for business stakeholders.

2. Database Overview

The Chinook database is a relational database implemented in PostgreSQL. It consists of 11 tables, which can be grouped into three main business areas:

- **Media Catalog (Product Data):**
  - track, album, artist, genre, media_type: Contains information about the music/video items sold.
  - playlist, playlist_track: Tracks how media is grouped.
- **Customer & Employee (CRM Data):**
  - customer: Contains customer demographics (country, city, email) and links to their support rep.
  - employee: Contains staff details, hierarchy (reports_to), and hire dates.

- **Sales (Transactional Data):**
    - invoice: Records the purchase event, date, billing address, and total amount.
    - invoice_line: The detail of each invoice (which track was bought, unit price, quantity).

Figure 1 Schema Chinook

3. Project Objectives

- **Data Preparation:** Consolidate and prepare data from multiple relational tables for seamless analysis.
- **Descriptive Analytics:** Explore sales trends over time, top-performing regions, and most popular music genres.
- **Business Strategy:** Deliver actionable recommendations for marketing focus, inventory expansion, and sales team optimization based on data evidence.

II. Data Cleansing & Transformation Plan

The dataset used in this analysis is relatively clean and well-structured, as it is derived from the Chinook database. However, several key data preparation and transformation steps were performed to ensure the dataset is suitable for meaningful business analysis. These steps help convert raw transactional data into a structured format that supports aggregation, comparison, and visualization.

Creation of Temporal Variables

To support time-based analysis, the invoice date was transformed to extract the year component. This transformation enables the evaluation of revenue trends over time, particularly for identifying year-over-year growth or decline in sales performance. By isolating the year variable, the dataset becomes more suitable for trend analysis and visualization.

Aggregation of Sales Data

To generate meaningful business insights, sales data was aggregated across several key dimensions, including billing country, year, and music genre. Aggregation allows large volumes of transactional data to be summarized into interpretable metrics, such as total revenue and total sales. This step is essential for identifying high-performing markets, analyzing sales trends over time, and understanding customer preferences across different genres.

These transformations ensure that the data is structured appropriately for business analysis and visualization.

III. Data Exploration

1. Market Expansion Strategy (Geographic Revenue Analysis)

1.1 Business Context:

Understanding geographic performance is crucial for allocating marketing budgets and identifying core markets. We used SQL to aggregate total sales by billing country. The resulting dataset was then passed to Python to visualize the revenue distribution, helping us pinpoint the most profitable regions

1.2 SQL Implementation:

```
query = """
SELECT billing_country, SUM(total) AS revenue
FROM invoice
GROUP BY billing_country
ORDER BY revenue DESC
"""

df_country = pd.read_sql_query(query, engine)
df_country.head()
```

	billing_country	revenue
0	USA	523.06
1	Canada	303.96
2	France	195.10
3	Brazil	190.10
4	Germany	156.48

.

Figure 2 Query Use for Market Expansion Strategy

1.3 Data Visualization:

```
import matplotlib.pyplot as plt

df_country.plot(
    x="billing_country",
    y="revenue",
    kind="bar",
    figsize=(10,5),
    title="Total Revenue by Country"
)

plt.show()
```

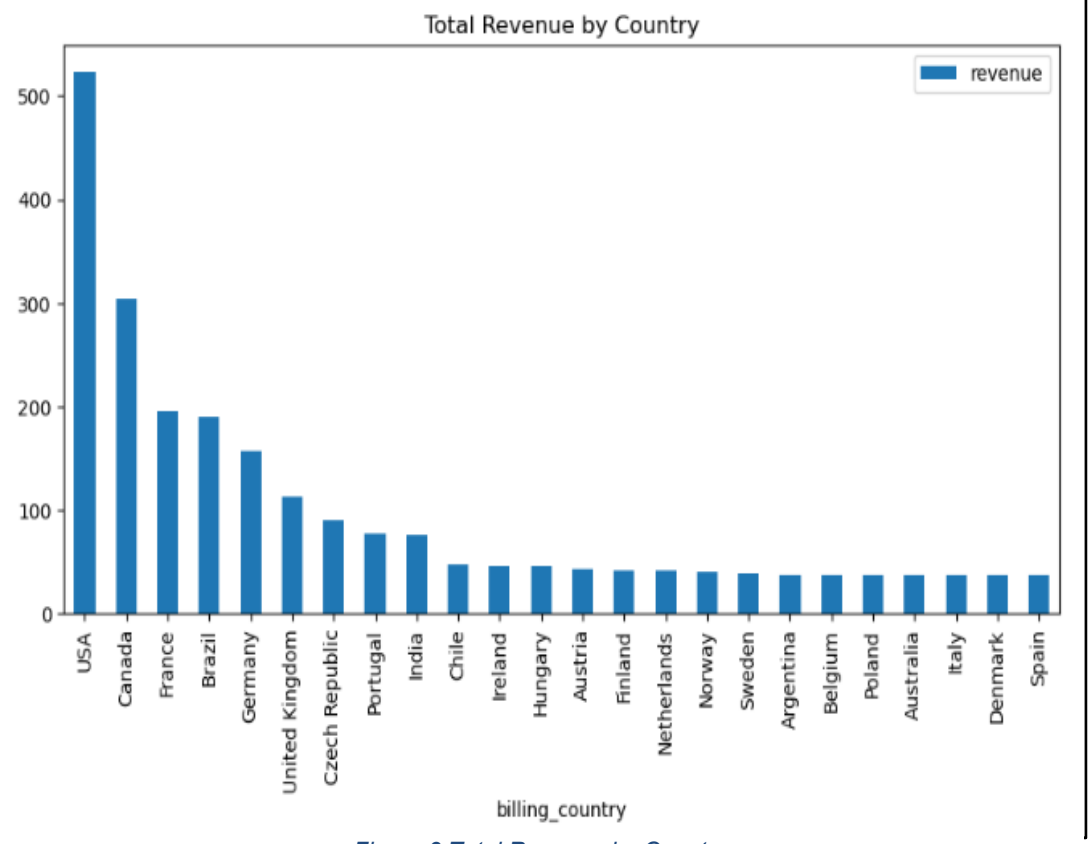

Figure 3 Total Revenue by Country

1.4 Insight & Business Implication:

- **Insight:** Figure 1 clearly illustrates that the United States is the dominant market, contributing 523.06 to the total revenue, followed by Canada (303.96) and France ($195.10).
- **Implication:** North America is the primary revenue driver for Chinook. Therefore, marketing efforts and customer retention strategies should be heavily focused on these key regions to maximize ROI (Return on Investment). Meanwhile, European countries like France and Germany show strong potential, suggesting that localized marketing campaigns in Europe could drive further growth.

2. Revenue Health Check (Year-over-Year Sales Trend)

2.1 Business Context:

To assess the overall business health, we analyzed the total revenue on a year-over-year basis to identify trends of growth or decline. We used SQL's EXTRACT function to isolate the year from the invoice timestamps, and then plotted a line chart in Python to observe the revenue trajectory from 2021 to 2025.

2.2 SQL Implementation:

```
query = """
SELECT EXTRACT(YEAR FROM invoice_date) AS year,
       SUM(total) AS revenue
FROM invoice
GROUP BY EXTRACT(YEAR FROM invoice_date)
ORDER BY year
"""

df_trend = pd.read_sql(query, engine)
print(df_trend)
df_trend["year"] = df_trend["year"].astype(int)
```

```
      year  revenue
0   2021.0   449.46
1   2022.0   481.45
2   2023.0   469.58
3   2024.0   477.53
4   2025.0   450.58
```

Figure 4 Query Use For Revenue Health Check

(Note: The 'year' column was converted to integer format using .astype(int) to remove decimal points. This data cleaning step ensures the x-axis on the chart displays clear, categorical years like 2021, 2022 instead of continuous floats like 2021.0)

2.3 Data Visualization:

```
df_trend.plot(
    x="year",
    y="revenue",
    kind="line",
    marker="o",
    figsize=(8,4),
    title="Sales Trend Over Time"
)

plt.show()
```

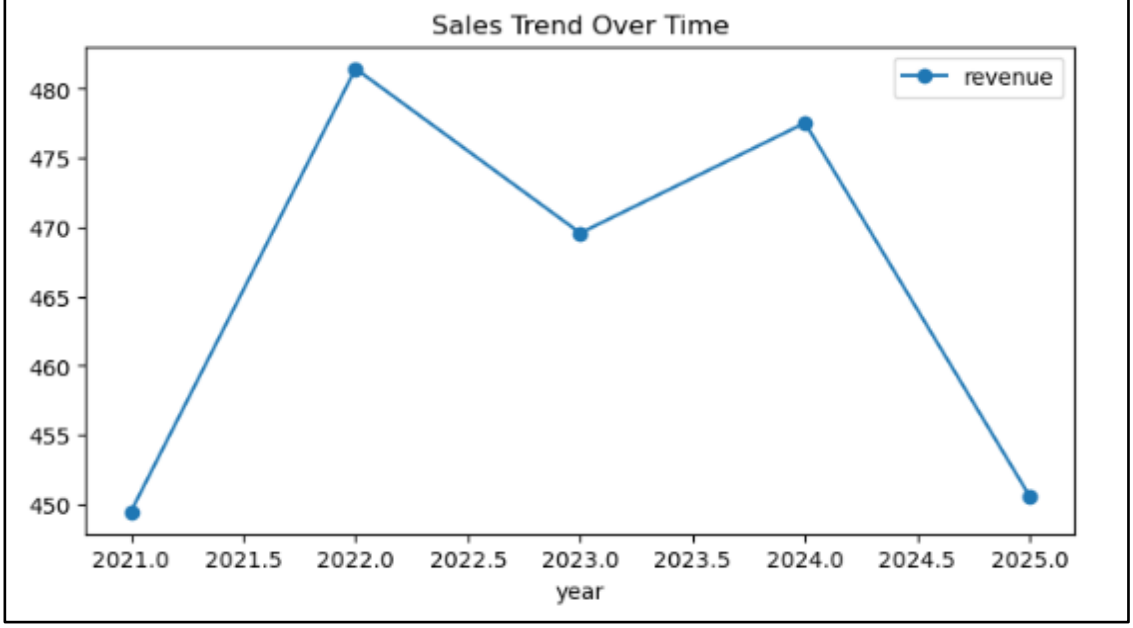

Figure 5 Sales Trend Over Ttime

2.4 Insight & Business Implication:

- **Insight:** Figure 2 demonstrates that the sales trend is highly fluctuating rather than showing steady growth. Revenue peaked in 2022 (481.45) and 2024 (477.53) but experienced noticeable drops in 2023 and especially in 2025 ($450.58).
- **Implication:** This volatility indicates that sales might be heavily dependent on new, blockbuster album releases rather than a consistent customer purchasing habit. To stabilize revenue, Chinook should consider introducing customer loyalty programs, promotional bundles during slow periods, or a monthly subscription service to maintain a steady cash flow.

3. Product Portfolio Optimization (Genre Popularity Analysis)

3.1 Business Context:

By analyzing sales performance across different music genres, we can identify customer preferences and optimize the digital music catalog. We used SQL JOIN statements to connect the tracks, genres, and invoice lines. The LIMIT 10 clause was applied to filter out low-selling categories and focus the Python visualization strictly on top-performing genres.

3.2 SQL Implementation:

```
query = """
SELECT g.name AS genre, SUM(il.quantity) AS total_sales
FROM genre g
JOIN track t ON g.genre_id = t.genre_id
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY g.name
ORDER BY total_sales DESC
LIMIT 10
"""

df_genre = pd.read_sql(query, engine)
print(df_genre)
```

```
                genre  total_sales
0                Rock          835
1               Latin          386
2               Metal          264
3  Alternative & Punk          244
4                Jazz           80
5               Blues           61
6            TV Shows           47
7           Classical           41
8            R&B/Soul           41
9              Reggae           30
```

Figure 6 Query Use For Product Portfolio Optimization

(Note: The LIMIT 10 clause was used in the SQL query to focus the analysis strictly on the top-performing genres, filtering out noise from low-selling categories.)

3.3 Data Visualization:

```
df_genre.plot(
    x="genre",
    y="total_sales",
    kind="bar",
    figsize=(10,5),
    title="Most Popular Music Genres"
)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

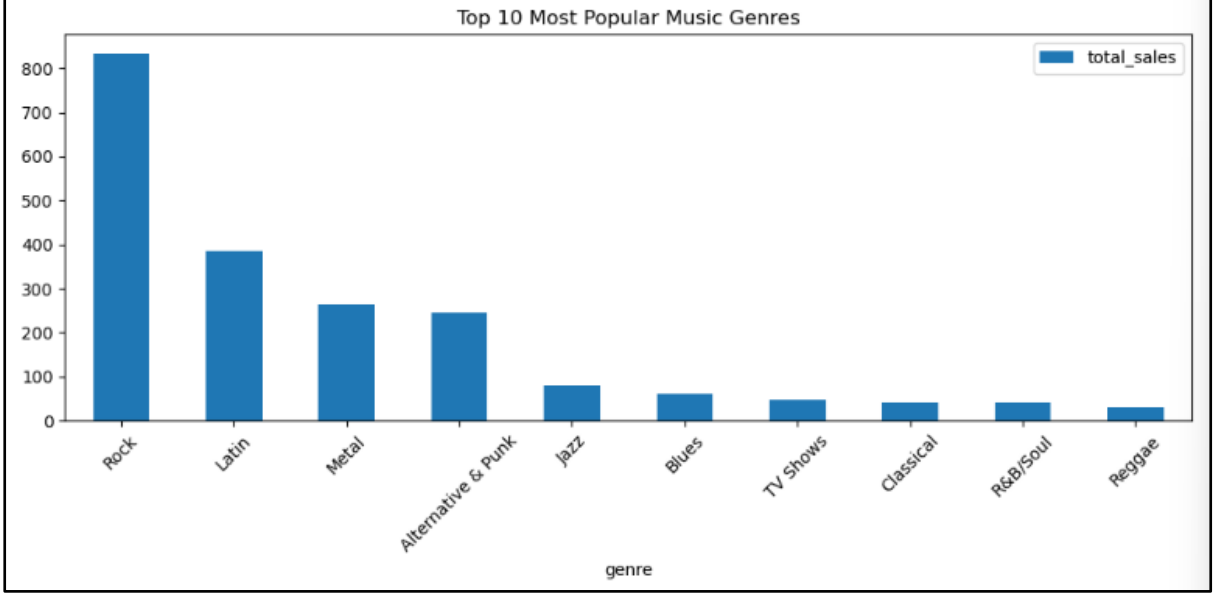

Figure 7 Top 10 Most Popular Music Genres

3.4 Insight & Business Implication:

- **Insight:** Figure 3 highlights that "Rock" is overwhelmingly the preferred genre, selling 835 tracks, which is more than double the second-place genre, "Latin" (386 tracks), followed by "Metal" (264 tracks).
- **Implication:** The Chinook customer base consists heavily of Rock and Metal fans. The licensing and purchasing team should prioritize acquiring a broader catalog of Rock and Alternative music. Additionally, targeted email campaigns featuring new Rock releases will likely yield the highest conversion rates.

IV. Conclusion & Strategic Recommendations

1. Executive Summary

Through the extraction of transactional data via PostgreSQL and visualization using Python, our analysis of the Chinook digital music store reveals three core business patterns:

- **Geographic Dominance:** North America is the undisputed primary profit center. The USA (523.06) and Canada (303.96) generate the highest revenue, far exceeding European markets.
- **Revenue Volatility:** The year-over-year sales trend (2021-2025) is highly fluctuating. The lack of steady, predictable growth suggests that Chinook's revenue heavily relies on sporadic blockbuster album releases rather than consistent customer purchasing habits.
- **Product Preferences:** "Rock" is the foundational pillar of Chinook's product portfolio, selling 835 tracks—more than double the runner-up genre ("Latin" with 386 tracks).

2. Strategic Recommendations

To maximize profitability, mitigate revenue risks, and ensure long-term business sustainability, we propose the following actionable strategies for Chinook's management team:

a. Revenue Stabilization via Subscription & Loyalty Programs

To address the fluctuating year-over-year sales, Chinook must shift from a purely transaction-based model to a recurring revenue model. Implementing a monthly music subscription tier or a "Chinook VIP" loyalty program (e.g., earning points for every track purchased) will help maintain a steady cash flow during months with fewer new album releases.

b. Focused Catalog Expansion & Licensing

Since Rock, Latin, and Metal account for the vast majority of sales, the purchasing and licensing team should allocate their budget accordingly. Prioritizing the acquisition of classic and trending Rock/Metal albums will satisfy the core customer base and yield the highest return on investment (ROI).

c. Geo-Targeted Marketing & Cross-Selling

Marketing budgets should be heavily concentrated on retaining high-value customers in the USA and Canada. Furthermore, Chinook can leverage the popularity of the "Latin" genre (the 2nd most popular) to run targeted promotional campaigns in South American markets (like Brazil, which is currently the 3rd largest single market) to drive further expansion. Less popular genres (like Jazz or Blues) can be bundled with Rock tracks at a discounted rate to clear digital inventory and encourage cross-genre listening.